Connor Williams
Assignment 5 - CSC 351

1. SQL Statement showing all books



```
1 •  SELECT * FROM books
```

100% 20:1

Result Grid | Filter Rows: Search | Export:

bookid	title	author	category	price
1	The Hobbit	J.R.R. Tolkien	Fantasy	15.99
2	1984	George Orwell	Dystopian	12.5
3	To Kill a Mockingbird	Harper Lee	Fiction	14.25
4	Dune	Frank Herbert	Sci-Fi	18
5	Foundation	Isaac Asimov	Sci-Fi	16.5
6	Pride and Prejudice	Jane Austen	Romance	10.99
7	The Great Gatsby	F. Scott Fitzgerald	Classic	11.5
8	Moby-Dick	Herman Melville	Adventure	13.99
9	War and Peace	Leo Tolstoy	Historical	19.99
10	The Odyssey	Homer	Classic	12.99
11	Crime and Punish…	Fyodor Dostoev…	Classic	14.5
12	The Catcher in the…	J.D. Salinger	Fiction	10.5
13	Brave New World	Aldous Huxley	Dystopian	13.5
14	The Lord of the Rings	J.R.R. Tolkien	Fantasy	25.99
15	Fahrenheit 451	Ray Bradbury	Dystopian	11.99
16	Jane Eyre	Charlotte Bronte	Romance	12
17	The Chronicles of…	C.S. Lewis	Fantasy	21.5
18	Animal Farm	George Orwell	Dystopian	9.99
19	The Alchemist	Paulo Coelho	Adventure	14
20	The Little Prince	Antoine de Saint…	Fable	9.5
21	Dracula	Bram Stoker	Horror	11.25
22	Frankenstein	Mary Shelley	Horror	10.75
23	The Picture of Dori…	Oscar Wilde	Classic	11
24	Les Miserables	Victor Hugo	Historical	18.5
25	The Count of Mont…	Alexandre Dumas	Adventure	16
26	The Brothers Kara…	Fyodor Dostoev…	Classic	17.5
27	Anna Karenina	Leo Tolstoy	Historical	15.5
28	Catch-22	Joseph Heller	Satire	13.25

books 1

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23	The Picture of Dori…	Oscar Wilde	Classic	11
24	Les Miserables	Victor Hugo	Historical	18.5
25	The Count of Mont…	Alexandre Dumas	Adventure	16
26	The Brothers Kara…	Fyodor Dostoev…	Classic	17.5
27	Anna Karenina	Leo Tolstoy	Historical	15.5
28	Catch-22	Joseph Heller	Satire	13.25
29	The Handmaid's Tale	Margaret Atwood	Dystopian	14.5
30	Neuromancer	William Gibson	Sci-Fi	15
31	Snow Crash	Neal Stephenson	Sci-Fi	16.25
32	Hyperion	Dan Simmons	Sci-Fi	17
33	Ender's Game	Orson Scott Card	Sci-Fi	14.75
34	I Robot	Isaac Asimov	Sci-Fi	13
35	The Martian	Andy Weir	Sci-Fi	15.5
36	The Silmarillion	J.R.R. Tolkien	Fantasy	18.99
37	American Gods	Neil Gaiman	Fantasy	16.5
38	Neverwhere	Neil Gaiman	Fantasy	13.5
39	Good Omens	Terry Pratchett	Fantasy	14.25
40	The Name of the…	Patrick Rothfuss	Fantasy	17.25
41	The Fellowship of t…	J.R.R. Tolkien	Fantasy	14.99
42	The Two Towers	J.R.R. Tolkien	Fantasy	14.99
43	The Return of the…	J.R.R. Tolkien	Fantasy	14.99
44	Lord of the Flies	William Golding	Classic	10.25
45	The Sun Also Rises	Ernest Hemingway	Classic	12.25
46	A Farewell to Arms	Ernest Hemingway	Classic	12.75
47	The Old Man and t…	Ernest Hemingway	Classic	9.75
48	The Grapes of Wrath	John Steinbeck	Historical	14.25
49	Of Mice and Men	John Steinbeck	Classic	10.5
50	East of Eden	John Steinbeck	Historical	16.75

2. Complete node.js server

server.js > ...

```
const express = require('express');
const pool = require('./mysql');

const app = express();
const port = 3000;

//GET
app.get('/', async (req, res) => {

    // Extract parameters from the URL
    const { category, author, format } = req.query;

    // Create the SQL query SELECT statement
    let sql = 'SELECT * FROM books WHERE 1=1';
    const params = [];

    // Search by category
    if (category) {
        sql += ' AND category = ?';
        params.push(category);
    }

    // Search by author
    if (author) {
        sql += ' AND author = ?';
        params.push(author);
    }

    try {
        //Execute
        const [books] = await pool.execute(sql, params);

        // Return HTML if text format is requested
        if (format === 'text') {
            let html = '<h1>Book Results</h1>';

            if (books.length === 0) {
                html += '<p>No books found.</p>';
            }

            books.forEach(book => {
                html += `
                    <p>
                        ${escapeHtml(book.title)} -
                        ${escapeHtml(book.author)} -
                        ${escapeHtml(book.category)} -
                        $${book.price}
                    </p>
```



```
                        ${book.price}
                    </p>
                `;
            });

            return res.type('html').send(html);
        }

        res.json(books);

    } catch (error) {
        // Error handling
        console.error('Database error:', error);

        res.status(500).json({
            error: 'Database query failed'
        });
    }
});

//Getting rid of special characters for the HTML
function escapeHtml(value) {
    return String(value).replace(/[&<>"']/g, char => ({
        '&': '&amp;',
        '<': '&lt;',
        '>': '&gt;',
        '"': '&quot;',
        "'": '&#39;'
    })[char]);
}

// Starting the server
app.listen(port, () => {
    console.log(`Server running at http://localhost:${port}`);
});
```

mysql.js (Password not included)

```
const mysql = require('mysql2/promise');

// Configuration details
const dbConfig = {
    host: "localhost",
    user: "root",
    password: "",
    database: "Assignment5",
    // Connection Pool settings are optional but good practice
    waitForConnections: true,
    connectionLimit: 10,
    queueLimit: 0
};

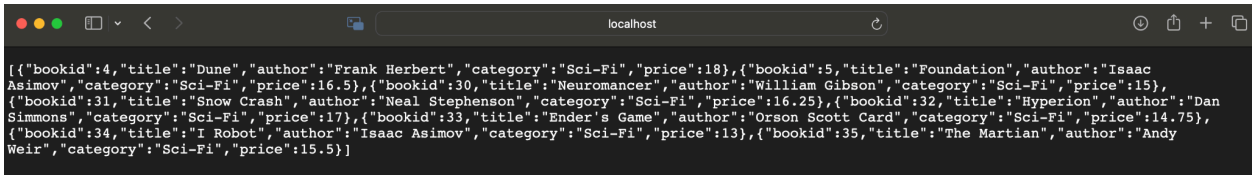
// Create and export the connection pool
const pool = mysql.createPool(dbConfig);

// A simple check to ensure the pool is working on startup
pool.getConnection()
    .then(connection => {
        console.log("Database Pool: MySQL connection successful and pool established.");
        connection.release(); // Release the test connection immediately
    })
    .catch(err => {
        console.error("Database Pool Error: Could not connect to MySQL! Check credentials and if the server is running.", err.message);
    });

module.exports = pool;
```

Examples:

http://localhost:3000/?category=Sci-Fi



http://localhost:3000/?author=George%20Orwell

